Aligning LLMs with Biomedical Knowledge using Balanced Fine-Tuning

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NEURAL INFORMATION PROCESSING SYSTEMS

Tencent AI for Life Sciences Lab

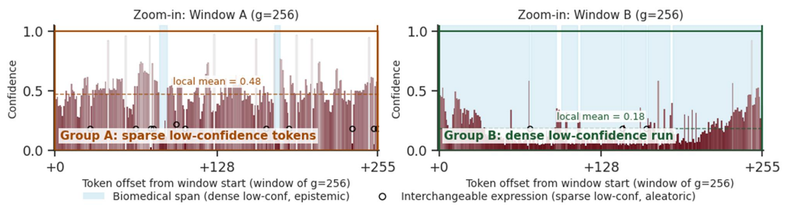
NeurIPS 2026 arxiv.org/abs/2511.21075
Project and code: shapsider.github.io/BFT

1 Challenge: learning unfamiliar science

LLMs need to learn unfamiliar gene names and biological mechanisms from limited expert supervision. Confidence-based training (DFT) reduces attention to difficult words, including those carrying useful scientific knowledge.

Difficult words can form meaningful passages

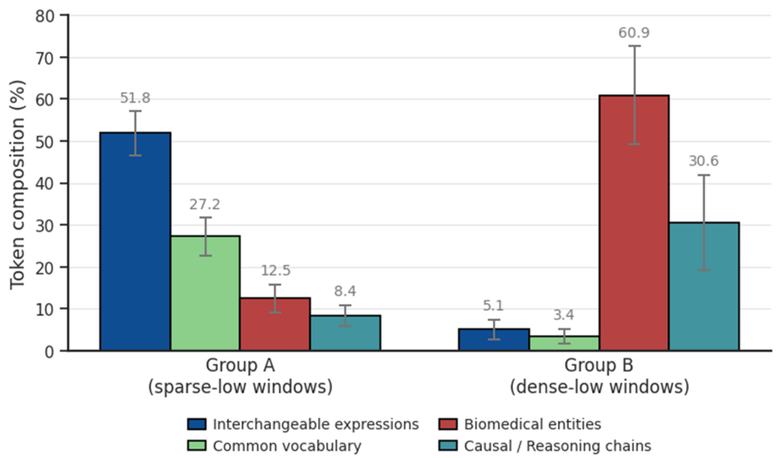
The model is uncertain about one word in an easy passage, or about many words in a difficult passage. These situations need different treatment.



Left: an isolated difficult word. Right: a whole difficult passage. A low prediction confidence alone cannot tell us which content the model needs to learn.

2 Motivation: preserve useful difficulty

The grouped difficult passages contain more biomedical entities and reasoning steps. Suppressing them can remove the very knowledge needed for domain adaptation.



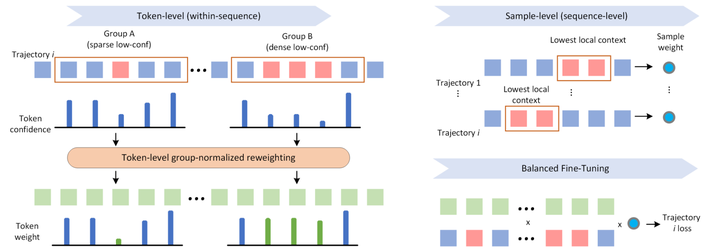
Biomedical entities rise from 12.5% in isolated-difficulty windows to 60.9% in grouped-difficulty windows. Causal and reasoning content rises from 8.4% to 30.6%. These are descriptive associations.

Goal: extending LLM capabilities to STEM

Learn scientific content while preserving broad knowledge. We test this in medicine and biology.

3 Method: Balanced Fine-Tuning

BFT changes how much attention each part of the training data receives. It uses the model's own confidence and nearby text, with two connected decisions.



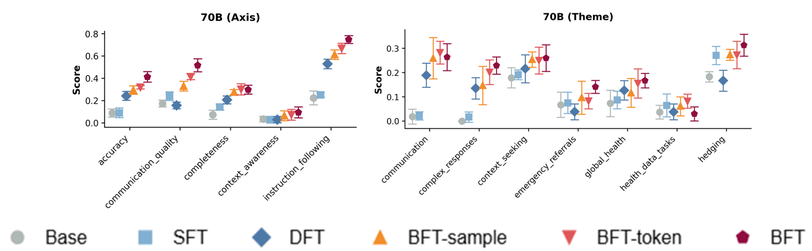
Left: compare each word with its surroundings, preserving difficult biomedical passages.
Right: give examples containing hard passages more attention, with a bounded contribution.

A small change to supervised training. No reward model or external uncertainty estimator is needed. One extra setting controls context size.

4 Results: biomedical reasoning

Matched training settings across 14B, 32B and 70B models. Original results below show medical competence and biological process reasoning.

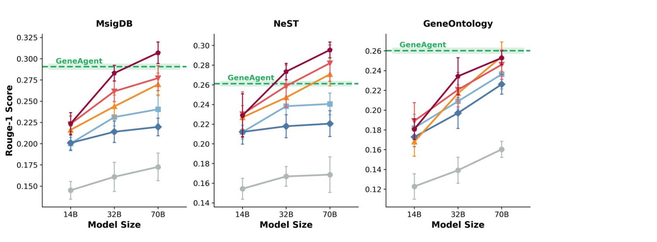
Medical competence: HealthBench Hard (70B)



BFT improves medical competence more consistently across model sizes. On VCWorld, BFT reaches 0.696 vs SFT 0.562, then 0.736 after feedback training.

Biological reasoning: GeneAgent

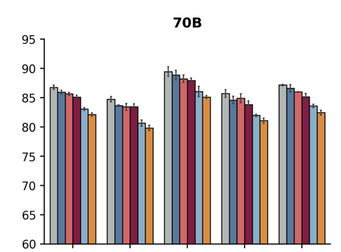
BFT has the best overall text-overlap score among the aligned open-weight models, with the agent pipeline held fixed.



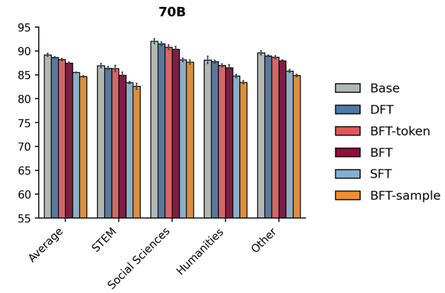
4 Results: STEM adaptation & retention

BFT expands medical and biological skills while retaining more broad knowledge than SFT. MMLU and CMMLU test broad knowledge across three model sizes. The 70B results appear below.

MMLU (70B)



CMMLU (70B)

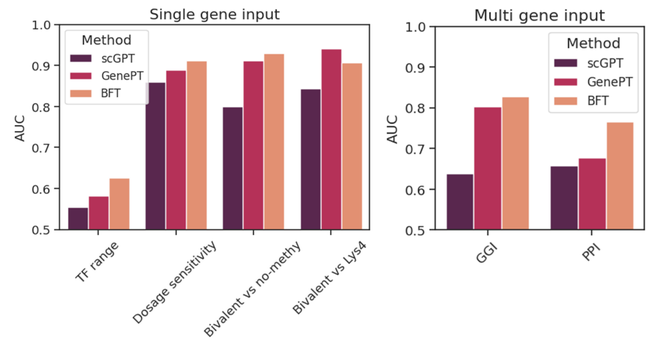


Categories in both plots: average, STEM, social sciences, humanities, other.

BFT retains more general capability than SFT and sample-only BFT. Some forgetting remains. DFT forgets least, while BFT offers stronger biomedical adaptation.

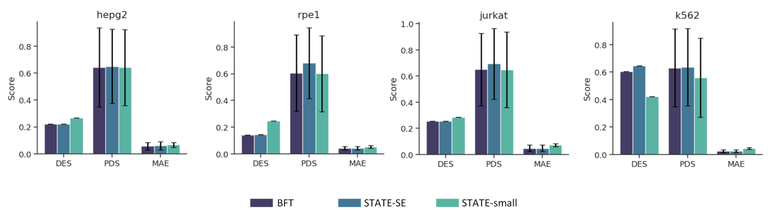
Gene descriptions support scientific prediction

BFT writes gene descriptions. A shared text encoder converts them into numerical representations for predicting gene properties and interactions.



Average gene prediction score: BFT 0.829 vs GenePT 0.802. GenePT remains higher on Lys4. Both use the same text encoder, with different source descriptions.

Single-cell response prediction



BFT profiles give comparable STATE prediction results across four datasets. Profile generation uses no training on these datasets. The STATE prediction model is still trained.